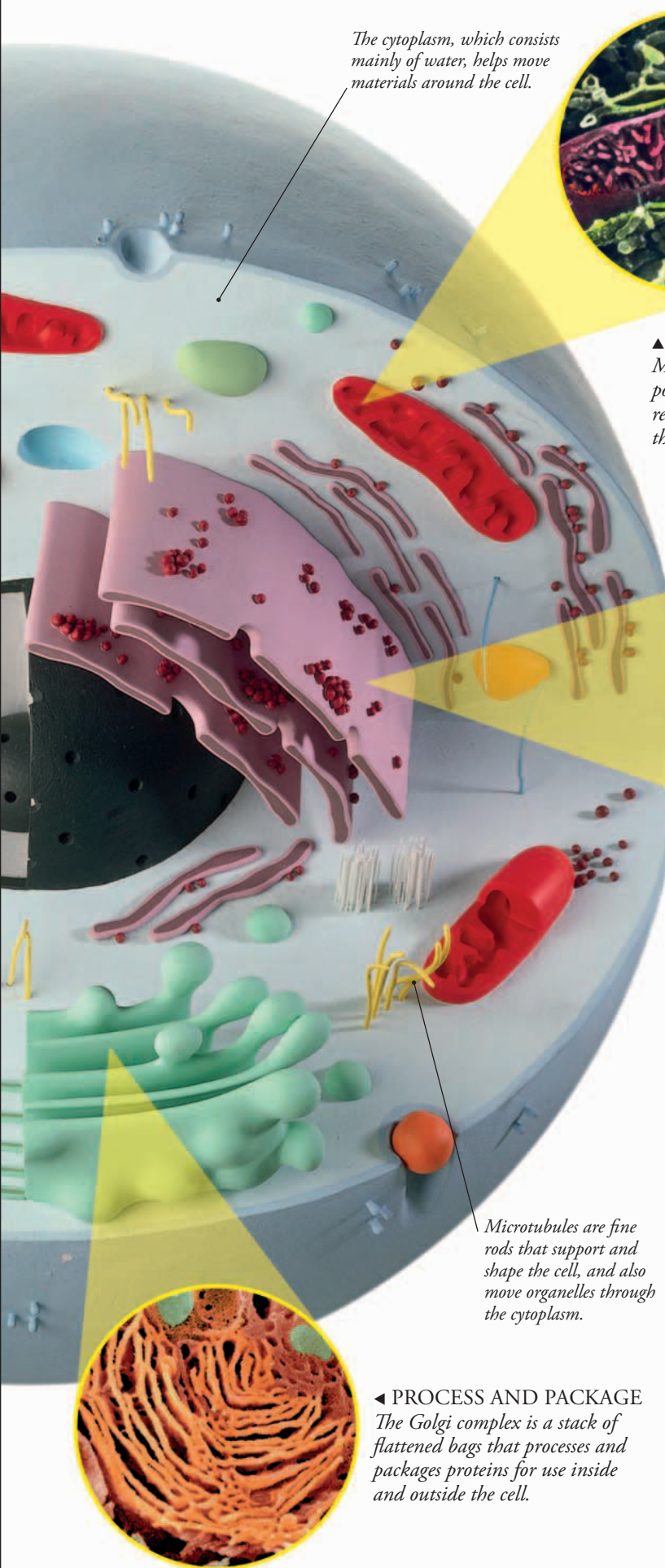
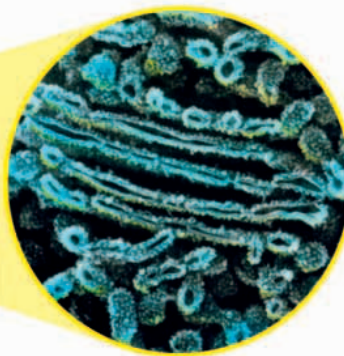




The cytoplasm, which consists mainly of water, helps move materials around the cell.

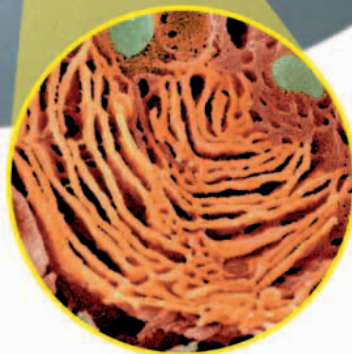


▲ **POWERHOUSES**
Mitochondria are the powerhouses of the cell, releasing energy to keep the cell alive.



◀ **PROTEIN FACTORIES**
This organelle, called the endoplasmic reticulum, is the busiest area of the cell. It helps to move and store proteins made by tiny ribosomes on its surface.

Microtubules are fine rods that support and shape the cell, and also move organelles through the cytoplasm.



◀ **PROCESS AND PACKAGE**
The Golgi complex is a stack of flattened bags that processes and packages proteins for use inside and outside the cell.

FAST FACTS

- Humans start life as a single cell, which divides and gives rise to the trillions of cells that make up a body.
- All living things—including animals and plants—are made of cells. Some, such as bacteria, consist of a single cell.
- The cells with the shortest lifespan are those lining the small intestine. They live for about 36 hours. Brain cells, however, may last a lifetime.



LOOK CLOSER: SEEING BIG

Invented in the 1930s, electron microscopes can magnify the insides of cells up to 200,000 times. Ordinary light microscopes have a maximum magnification of just 2,000 times, so they reveal much less detail.



▲ **ELECTRON MICROSCOPE** *A scientist looks at a 3-D image created by a scanning electron microscope. The same image can be viewed on the monitor's screen.*